Visual Telemetry: Temporal State Estimation in MOBA Gameplay via Deep Residual Networks

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December 2025

Abstract

This project addresses the task of predicting game progress in League of Legends (LoL) matches from single frame screenshots using computer vision techniques. We employ three substantial components as required by the course scope: **Dataset Creation** by curating high-ELO gameplay footage, **Fine-Tuning** a pre-trained ResNet-50 model, and **Hyperparameter Tuning** through targeted experimental sweeps. The dataset consists of 7,369 frames extracted at 5-second intervals from 19 videos. We formulate the problem as regression (continuous time), fine-grained classification (7 bins), and coarse-grained classification (3 phases). Results demonstrate that **Coarse-Grained Classification** achieves **86.07%** accuracy, confirming the model effectively learns distinct game phases. **Regression** achieves a mean absolute error of **156.03 seconds** (2.6 minutes). However, **Fine-Grained Classification** plateaued at **49.53%**, highlighting the visual ambiguity of short temporal intervals. Analysis reveals that performance is highly stable in the early-to-mid game but degrades significantly in late game scenarios due to data scarcity.

1 Introduction

League of Legends is a complex MOBA where game state correlates strongly with time. This project investigates whether a deep learning model can predict game time solely from visual information (minimap state, structures, champion stats) without reading the on-screen timer.

We investigate three formulations:

- **Regression**: Continuous time prediction.
- **Fine-grained Classification**: 7 classes (5-minute intervals).
- **Coarse-grained Classification**: 3 classes (Early/Mid/Late Game).

1.1 Scope of Work

This project addresses three major components:

1. **Dataset Creation**: Curated 19 videos (11 Train, 4 Val, 4 Test) across 4 roles, totaling 7,369 images.
2. **Fine-Tuning**: Adapted ImageNet-pretrained ResNet-50 for game state understanding.
3. **Hyperparameter Tuning**: Systematic experiments with learning rates ($1e^{-4}, 3e^{-4}, 1e^{-3}$) for all three tasks.

2 Methods

2.1 Dataset Curation

We constructed a custom dataset sourced from YouTube gameplay videos. To ensure data quality and difficulty, we applied strict inclusion criteria:

- **Source**: High-ELO (Challenger) Korean professional players.
- **Roles**: Top, Jungle, Mid, and ADC. Support was excluded as it shares lane perspective with ADC, offering redundant visual information.
- **Video Duration**: Ranging from 20 to 40 minutes to capture diverse game lengths.

Frames were extracted at 5-second intervals using FFmpeg. The train/val/test split (11/4/4 videos) was stratified by role to ensure balanced representation across all splits.

2.1.1 Data Distribution

The final dataset comprises **7,369 frames** extracted from **19 unique videos**. As shown in Table 1, the data follows a natural long-tail distribution, with fewer samples available for late game states (25+ min) compared to the early game.

Coarse Class	**Count**	**%**
Early Game (0-14 min)	3,154	42.8%
Mid Game (14-25 min)	2,506	34.0%
Late Game (25+ min)	1,709	23.2%
Total	**7,369**	**100%**

Fine Bin	**Count**	**%**
0-5 min	1,102	15.0%
5-10 min	1,140	15.5%
10-15 min	1,140	15.5%
15-20 min	1,140	15.5%
20-25 min	1,138	15.4%
25-30 min	920	12.5%
30+ min	789	10.7%
Total	**7,369**	**100%**

Table 1: Dataset Distribution. Coarse and fine-grained bin boundaries do not align (10-15 min spans both Early and Mid Game).

2.1.2 Preprocessing and Anti-Leakage Strategy

To ensure the model learns game state rather than performing OCR (Optical Character Recognition), we implemented a masking strategy:

- **Timer & KDA**: The top-center game timer and bottom-center scoreboard were masked.
- **Noise Handling**: Non-gameplay elements (”Subscribe for more!” overlays) were retained to test robustness.

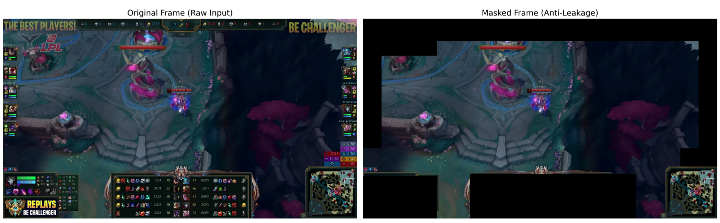


Figure 1: Left: Raw frame with timer visible. Right: Masked frame used for training

2.2 Training Configuration

We used a ResNet-50 backbone with the following configuration:

- **Optimizer**: AdamW (Weight Decay $1e^{-4}$)
- **Batch Size**: 32
- **Epochs**: 25
- **Scheduler**: ReduceLROnPlateau (Patience=3)
- **Augmentation**: ColorJitter (Brightness/Contrast 0.15). *Note: Geometric augmentations like flipping were disabled due to the asymmetrical nature of the MOBA map.*

3 Results and Analysis

3.1 Hyperparameter Tuning

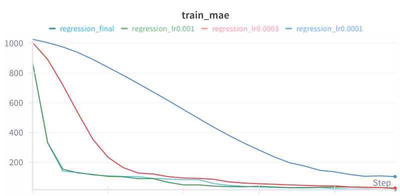
We performed a targeted sweep to select the optimal Learning Rate (LR) based on Validation Set performance.

Task	**LR=1e-4**	**LR=3e-4**	**LR=1e-3**
Regression (Val MAE ↓)	131.30s	128.80s	**118.88s**
Fine Bins (Val Acc ↑)	0.556	0.581	**0.593**
Coarse Bins (Val Acc ↑)	0.860	0.878	**0.921**

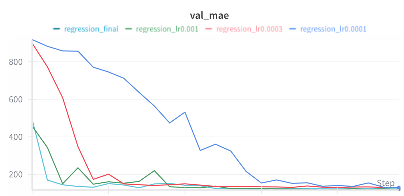
Table 2: Validation metrics for Learning Rate selection.

3.2 Training Dynamics

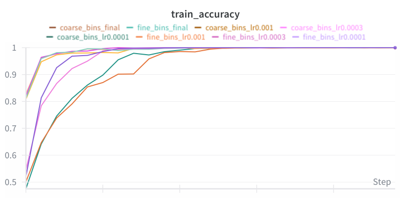
Figure 2 illustrates the training progression. While the MAE decreases steadily, the Accuracy curves reveal the stability of the model.



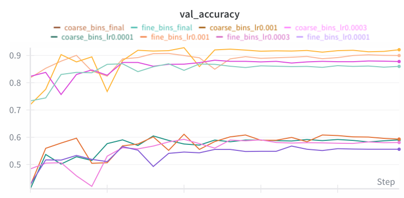
(a) Train MAE



(b) Validation MAE



(c) Train Accuracy



(d) Validation Accuracy

Figure 2: Training and Validation Metrics. The regression model shows consistent convergence, while the classification accuracy highlights the difficulty ceiling for fine-grained tasks.

3.3 Test Set Performance

Performance on the Test Set is summarized in Table 3.

Category	Regression (MAE)	Fine Bins (Acc)	Coarse Bins (Acc)
Random Baseline	507.40s	14.29%	33.33%
Overall	**156.03s**	**49.53%**	**86.07%**
Top Lane	143.98s	54.8%	93.0%
Jungle	124.72s	54.5%	90.1%
Mid Lane	220.25s	36.8%	73.8%
ADC	121.29s	54.6%	89.5%
Early Game (0-14 min)	106.73s	59.6%	98.8%
Mid Game (14-25 min)	127.10s	47.9%	91.9%
Late Game (25+ min)	311.92s	30.5%	48.7%

Table 3: Detailed Test Set Performance breakdown by Role and Game Phase. Classification baselines assume uniform random prediction, regression baseline predicts the training set mean.

3.3.1 Analysis of Late Game Performance

A key observation from Table 3 is the sharp decline in metrics for the Late Game phase (25+ minutes). While the Coarse model achieves near-perfect accuracy in the Early Game (98.8%) and strong accuracy in the Mid Game (91.9%), the overall average is heavily penalized by the Late Game drop (48.7%).

This degradation correlates directly with the dataset distribution (Table 1), where the 30+ minute bin contains only 10.7% of the data. This data scarcity creates a bottleneck for the model in learning the nuanced visual features of late-game states.

3.3.2 Regression Analysis

The regression model achieved a Test MAE of 156.03 seconds. Figure 3 shows the predictions vs. actual timestamps. Note the increased variance in the upper right quadrant (25+ minutes), consistent with the long-tail data distribution issue.

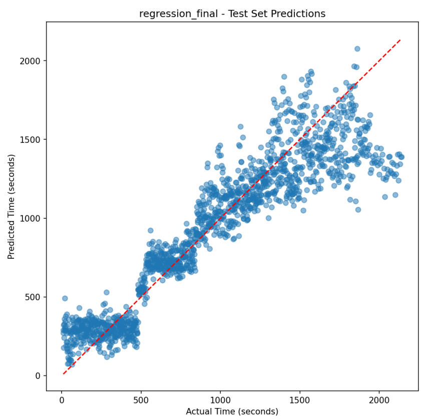
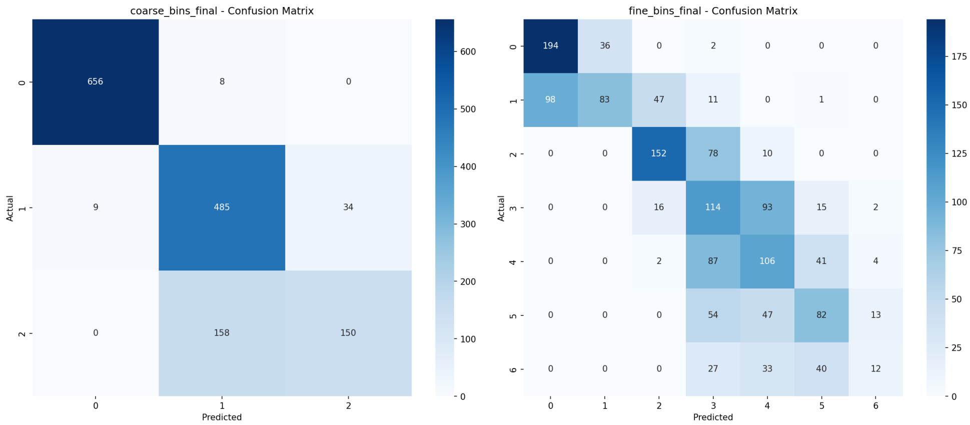


Figure 3: Regression Predictions. The variance increases significantly in the late game (25+ min) due to limited training samples.

3.3.3 Classification Confusion Matrices

Figure 4 visualizes the classification errors. The Coarse model shows strong performance in the early stages, whereas the Fine model (Right) shows confusion between adjacent bins, which is expected given the visual similarity of 5-minute intervals.



(a) Coarse Bins (86.1% Acc)

(b) Fine Bins (49.5% Acc)

Figure 4: Confusion Matrices. The Coarse model (Left) accurately identifies early game but struggles to distinguish mid-game from late-game states. The Fine model (Right) exhibits a tight distribution around the diagonal, suggesting that misclassifications are mostly "near-misses", reflecting the continuous nature of game progression.

3.4 Visual Interpretability

To validate that the model learns semantic game features rather than relying on shortcuts, we applied Grad-CAM analysis to the best-performing model (Coarse Bins). Figure 5 illustrates the attention maps for test samples. The visualizations confirm that the model attends to semantically relevant areas, such as the **Minimap** and **Champion Clusters** rather than overfitting to static UI elements.

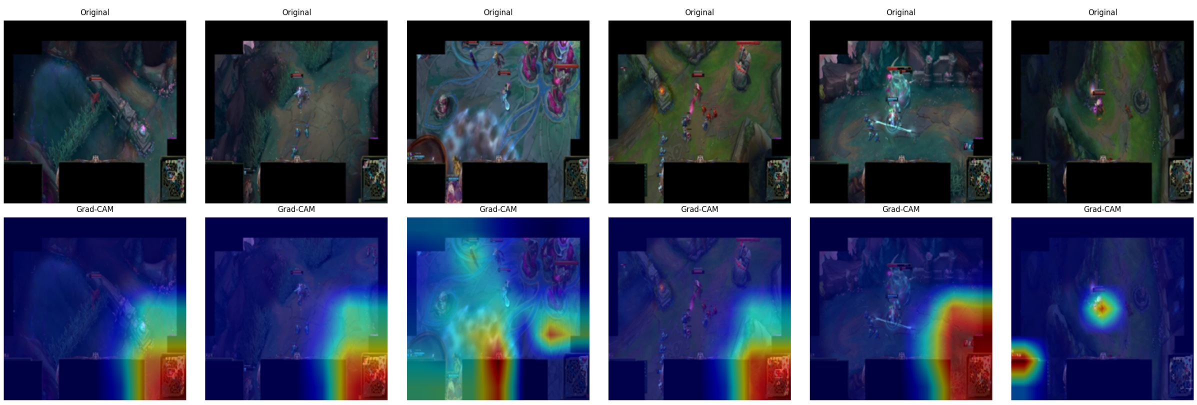


Figure 5: Grad-CAM visualizations for the Coarse Classification model. The heatmaps highlight active gameplay areas (champions, effects) and the minimap state, indicating effective feature learning without reliance on static HUD statistics.

4 Conclusion and Future Work

We successfully developed a vision-based game progress predictor, achieving **86.07%** coarse-grained accuracy and a regression MAE of **156.03 seconds** (2.6 minutes). The effective use of masking confirmed that the model relies on genuine game state semantics (structure health, map control) rather than OCR shortcuts. However, performance degradation in the late game (25+ minutes) identifies a critical bottleneck driven by the long-tail distribution of match durations.

Future Work: To address the data scarcity in late-game scenarios, future iterations will prioritize targeted collection of long-duration VODs and the incorporation of temporal architectures (LSTMs) to leverage sequential context between frames.

Supplementary Materials

The full training logs, metrics, and interactive visualizations are available on Weights & Biases.